

BUDDY: An Intelligent Robot Dog for Offline Voice Interaction, Visual Perception, and Safety-Aware Multimodal Control

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Abstract—Companion robots must integrate natural interaction, environmental perception, visual awareness, and physical actuation while operating safely around people. A critical systems challenge arises when high-level interaction components are connected to physical movement, because a recognized or generated command should not directly obtain authority over robot actuators.

This paper presents BUDDY, an intelligent multimodal robot-dog prototype integrating offline voice interaction, active visual perception, environmental sensing, and four-wheel mobility using a Raspberry Pi 4 and ROS 2. The system incorporates three research streams: offline auditory interaction, visual and facial-expression intelligence, and environment-aware navigation and physical control.

A principal architectural contribution is the separation of high-level movement intent from physical motor authority. High-level components publish movement requests rather than direct motor commands. An independent safety supervisor evaluates current ultrasonic observations before safety-approved commands are published to the motor executor and continuously reevaluates active movement when environmental conditions change.

Physical prototype validation demonstrated automatic startup, offline wake-word interaction, spoken forward, backward, left, right, and stop control, four-direction ultrasonic sensing, dynamic interruption of active forward movement when an obstacle was introduced, movement resumption after obstacle removal, and camera-based face following using a two-axis actuated head.

Index Terms—companion robotics, human–robot interaction, offline speech recognition, ROS 2, multimodal perception, safety-aware control

I. INTRODUCTION

Companion robots combine physical embodiment with natural human–robot interaction and are increasingly investigated for applications involving social engagement, assistance, education, and long-term interaction [1], [2]. Unlike conventional software assistants, an embodied companion must coordinate

perception, interaction, decision-making, and physical actuation while operating safely within a shared environment.

A dog-inspired companion robot may be expected to respond when called, interpret spoken instructions, observe environmental obstacles, orient toward a nearby person, observe facial cues, and move within its surroundings. These behaviours require integration of auditory processing, environmental sensing, visual perception, decision-making, navigation, and actuator control.

Implementing these capabilities together on embedded hardware creates additional constraints involving computation, memory, peripheral interfaces, electrical power, sensor reliability, and network availability.

Speech provides a particularly natural interaction mechanism. Although modern automatic speech-recognition systems such as Whisper demonstrate strong performance [3], continuous cloud dependence can introduce network dependency and privacy concerns. Local processing is therefore useful for fundamental companion functions, particularly commands that influence physical behaviour.

However, local speech processing alone does not address the safety problem created when high-level interaction is connected to physical actuation. A recognized movement request may be incorrect, the environment may change after the request begins, or sensor information may become unavailable.

Allowing conversational or high-level AI components to directly control motor hardware would therefore couple uncertain interpretation with safety-critical physical authority.

This paper presents BUDDY, an intelligent multimodal robot-dog prototype implemented on a Raspberry Pi 4 using ROS 2 [4]. BUDDY integrates offline auditory interaction, four-wheel mobility, four-direction ultrasonic perception,

camera-based visual perception, facial-expression processing, and a two-axis actuated head.

The central research question is:

How can multimodal human–robot interaction be integrated with physical movement on embedded companion-robot hardware while preventing high-level interaction components from obtaining unrestricted motor authority?

BUDDY addresses this question through a safety-separated control architecture. High-level components generate movement requests while an independent ROS 2 safety supervisor determines whether those requests may reach the physical motor executor. Environmental conditions continue to be reevaluated during active movement.

The principal contributions are:

- an integrated companion-robot prototype combining off-line auditory interaction, environmental sensing, physical mobility, active visual perception, and extensible autonomous behaviour;
- a modular ROS 2 architecture separating perception, high-level intelligence, safety validation, and physical actuation;
- a safety-supervised movement pipeline with continuous environmental reevaluation and fail-safe handling of stale obstacle information; and
- physical validation demonstrating automatic startup, off-line voice-controlled movement, four-direction obstacle sensing, dynamic interruption of active movement, movement resumption after obstacle removal, and camera-based face following.

II. RELATED WORK

Companion and socially interactive robots combine physical embodiment with interaction capabilities intended to support sustained engagement with human users.

Ahmed *et al.* reviewed human–robot companionship and identified anthropomorphic and zoomorphic embodiments as important approaches while highlighting opportunities for greater physical mobility [1].

Socially assistive robotics research similarly emphasizes that embodiment, autonomy, sensing, interaction, and actuation should be considered together when robots operate in human environments [2], [5].

Speech provides a natural interaction channel for companion robots. Large-scale automatic speech-recognition systems such as Whisper have demonstrated robust recognition across diverse speech conditions [3]. However, embedded companion systems may benefit from local recognition for basic interaction where continuous network availability cannot be assumed.

Visual perception provides a complementary interaction modality. DeepFace demonstrated learned representations for face verification [6], while FaceNet introduced an embedding space for face recognition and clustering [7]. Facial-expression recognition has also been extensively investigated using deep-learning approaches [8].

Within an embodied companion robot, visual perception can additionally influence physical behaviour such as orienting

a camera toward a detected user. However, isolated facial-expression predictions should not automatically be treated as direct measurements of a user’s internal emotional state.

ROS 2 provides middleware for modular robotic systems and supports communication between independently developed components [4]. BUDDY uses this modularity to integrate auditory, visual, environmental, and actuation components while maintaining a separate physical safety boundary.

Existing research establishes the individual foundations of speech processing, facial perception, social robotics, environmental sensing, and robot middleware. The systems-level challenge addressed by BUDDY is coordinating these capabilities within one embedded companion robot while separating uncertain high-level interaction from physical movement authority.

III. BUDDY SYSTEM DESIGN

BUDDY is organized into three principal layers: a hardware layer responsible for physical sensing and actuation, a software and intelligence layer responsible for perception and decision logic, and an application/control layer responsible for composing these outputs into observable robotic behaviour.

This organization separates the components that determine *what* BUDDY should do from the hardware interfaces that determine *how* physical actuation is performed.



Fig. 1. Physical BUDDY robot-dog prototype integrating mobile actuation, ultrasonic sensing, active camera-head control, auditory input, and embedded computation.

A. Hardware Layer

BUDDY uses a Raspberry Pi 4 Model B with 4 GB RAM as the principal embedded computer. The prototype operates using a 64-bit Ubuntu environment with ROS 2 Jazzy.

Physical mobility is provided by four DC geared motors controlled by two TB6612FNG dual H-bridge motor drivers. Four HC-SR04 ultrasonic sensors provide front-left, front-right, rear-left, and rear-right environmental observations.

Visual sensing is provided by an IMX219 CSI camera mounted on a two-axis actuated head. Two servo motors produce pan and tilt movement through a PCA9685 PWM controller.

An AB13X USB microphone provides local speech acquisition.

The Raspberry Pi computing supply is separated from the higher-current motor and servo supply. The actuation subsystem uses three protected 18650 cells with battery management, fuse protection, switching, and DC–DC regulation. A common electrical ground is maintained between computing and actuation subsystems.

Fig. ?? illustrates the principal hardware interconnections.

TABLE I
PRINCIPAL BUDDY PROTOTYPE COMPONENTS

Component	Function
Raspberry Pi 4B, 4 GB	Embedded computation and ROS 2
2× TB6612FNG	Four-motor drive control
4× DC geared motors	Physical mobility
4× HC-SR04	Directional obstacle sensing
IMX219 CSI camera	Visual sensing
2× servos + PCA9685	Pan-tilt head actuation
AB13X USB microphone	Local speech acquisition
Vosk	Offline speech recognition
Piper	Offline speech synthesis

B. Software and Intelligence Layer

The software layer consumes raw audio, camera images, and ultrasonic observations and produces interpreted perception outputs and high-level requests.

Physical movement requests are only released to hardware after passing through the safety architecture described in Section IV.

1) *Offline Auditory Interaction:* The auditory pipeline captures two-channel PCM audio at 48 kHz through the USB microphone. The channels are combined and converted to a 16 kHz mono stream before offline recognition using Vosk.

BUDDY uses the wake phrase “Buddy.” Following activation, recognized physical movement utterances are mapped to a constrained movement request set.

The physically validated spoken command vocabulary includes forward, backward, left, right, and stop.

Safety-relevant commands are handled deterministically. Conversational processing remains separate from direct motor actuation.

Local spoken feedback is generated using Piper with the `en_US-lessac-medium.onnx` voice model.

2) *Visual and Facial Intelligence:* The visual-perception component receives images from the IMX219 camera. During physical face-tracking experiments, the detected face position was compared with the camera-frame centre to determine a pan/tilt offset.

The offset is passed to the head-control component rather than being applied directly to servo hardware.

A seven-class convolutional neural network was also developed for angry, disgust, fear, happy, neutral, sad, and surprise expression categories.

The model receives 48×48 grayscale face images and contains three convolutional stages followed by fully connected classification layers.

The broader visual-affect research direction extends the classifier toward interaction using prediction confidence, temporal consistency, recognized-user context, and previous observations.

The facial-expression classifier is reported as a developed component. Complete real-time camera-to-expression-to-personalized-response operation was not considered physically validated during the reported experiments.

3) *Navigation and Environmental Intelligence:* Four directional ultrasonic observations provide environmental information to the navigation and safety subsystems.

The current physical implementation supports obstacle-aware movement intervention.

The broader navigation research direction extends toward autonomous indoor movement, patrol-state management, localization, route execution, and security-event behaviour.

Complete autonomous patrol is not reported as a validated result in the current evaluation.

C. Application Layer

The application layer connects software decisions with the appropriate physical actuator.

The two implemented perception-to-actuation branches are:

- **Voice branch:** microphone → offline speech recognition → movement request → safety supervisor → motor executor → motor drivers;
- **Vision branch:** camera → face detection → pan/tilt offset → head-control request → PCA9685 → pan-tilt servos.

In both cases, sensing hardware does not directly command physical actuation.

IV. SAFETY-AWARE MOTION ARCHITECTURE

A central system constraint is that high-level interaction software cannot directly authorize physical movement.

The principal control flow is:

High-Level Request → Safety Supervisor → Safe Movement
→ Motor Executor.

A. Voice-to-Motion Separation

Recognized movement instructions are represented as structured movement requests and published through:

`/movement/request`

The safety supervisor evaluates each request together with the current environmental state.

Approved movement is published through:

`/movement/safe`

The motor executor subscribes only to the safety-approved command topic.

This creates an explicit distinction between high-level movement intent and physical motor authorization.

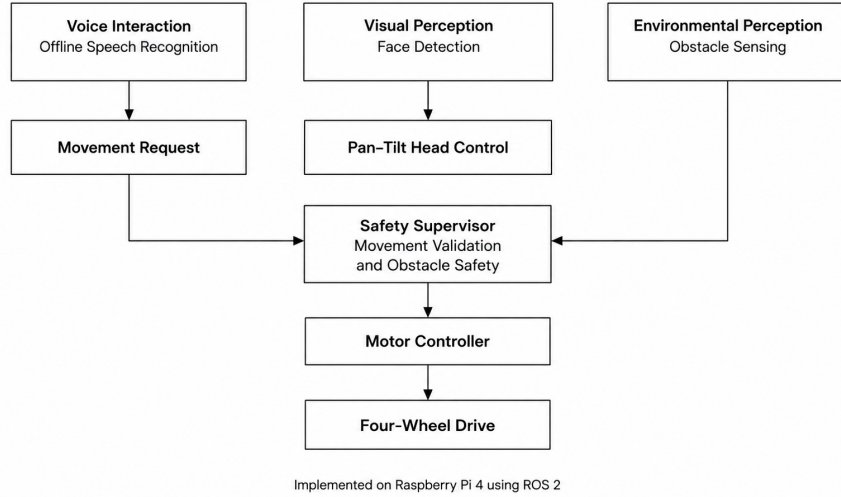


Fig. 2. System architecture of BUDDY showing the voice, visual, environmental perception, safety supervision, and physical actuation pathways.

B. Four-Direction Environmental Perception

The four HC-SR04 sensors observe the front-left, front-right, rear-left, and rear-right regions.

The prototype GPIO assignment is shown in Table II.

TABLE II
ULTRASONIC SENSOR GPIO CONFIGURATION

Sensor	TRIG	ECHO
Front-left	21	22
Front-right	23	24
Rear-left	25	26
Rear-right	27	14

The sensors are triggered sequentially with approximately 60 ms separation to reduce ultrasonic cross-interference.

Recent observations are filtered before being supplied to the safety logic.

The configured caution threshold is approximately 0.45 m and the stop threshold is approximately 0.25 m. These are system configuration parameters rather than experimentally derived ranging-accuracy values.

C. Dynamic Safety Reevaluation

Safety is evaluated continuously while a movement request remains active.

When obstacle information changes, the currently active movement request is reconsidered. This allows physical movement to be interrupted even when the original request was considered safe when it began.

Environmental information older than approximately 1.0 s is treated as stale.

During active movement, obstacle-state freshness is checked at approximately 0.1 s intervals.

If the environmental state is no longer trustworthy, the system transitions to STOP rather than assuming that continued movement remains safe.

D. Motor Execution and Hardware Isolation

The physical motor executor receives movement commands only from `/movement/safe`.

A separate approximately 0.05 s watchdog provides an additional defensive mechanism at the motor-execution layer.

The hardware abstraction layer separates ROS 2 control software from direct GPIO operations and supports controlled initialization, physical hardware execution, and cleanup.

This architecture does not constitute formal safety certification. Its purpose is to prevent uncertain high-level interaction components from directly obtaining unrestricted authority over the physical motors.

V. EXPERIMENTAL EVALUATION

A. Evaluation Procedure

The prototype was evaluated through physical integration experiments and software-level verification.

The evaluation focused on:

- automatic embedded-system startup;
- offline voice-controlled movement;
- four-direction environmental sensing;
- dynamic obstacle intervention;
- active visual face following; and
- movement and safety software verification.

Components that had not reached complete physical integration were excluded from claims of end-to-end validation.

B. Wheel-Speed Calibration

The drive subsystem was calibrated directly on the physical platform.

A normalized drive level of 0.1 produced movement that was too slow for practical operation, while 0.5 resulted in movement that was too rapid for controlled indoor use.

A drive level of approximately 0.2 was selected for straight movement. Turning required a higher operating level because lower values produced unstable rotation.

These values were used as prototype operating parameters in the subsequent physical experiments.

C. Automatic Startup and Voice Interaction

A cold-boot experiment evaluated whether BUDDY could reach an interactive state without manually launching the principal runtime components.

The Raspberry Pi was shut down and its computing power source disconnected.

Power was restored without establishing an SSH session.

Following operating-system startup, the BUDDY runtime initialized automatically and produced the spoken message:

“Hi there. Buddy is ready.”

The wake phrase “Buddy” was subsequently presented through the local microphone.

The system responded with “Yes?”, confirming that the local auditory pipeline was available after automatic startup.

Forward, backward, left, right, and stop commands were then evaluated through the complete implemented control path.

Each recognized physical command was converted into a movement request, evaluated by the safety supervisor, and delivered to the motor executor only after safety authorization.

Corresponding physical wheel behaviour was observed for all five movement commands.

A controlled multi-participant speech-recognition experiment was not completed; therefore, generalized command-recognition accuracy is not reported.

D. Four-Direction Environmental Perception

The front-left, front-right, rear-left, and rear-right ultrasonic channels were individually exercised to verify that observations from all four sensing positions were available to the environmental perception subsystem.

This experiment verified sensor integration and directional observation availability.

It was not designed as a calibrated ranging-accuracy experiment, and unsupported distance-error statistics are therefore not reported.

E. Dynamic Obstacle Experiment

The principal safety experiment evaluated whether an already-active movement request could be interrupted when the physical environment changed.

BUDDY was initially commanded to move forward while the front sensing region was clear.

After forward movement began, a physical obstacle was introduced into the front ultrasonic sensing region.

The updated obstacle observation caused the still-active movement request to be reevaluated by the safety supervisor.

A STOP command was delivered to the motor executor and physical forward movement ceased.

The obstacle was subsequently removed while the forward request remained active.

Once the environmental state returned to a permitted condition, the request was reconsidered and physical forward movement resumed.

F. Active Face-Following Experiment

Active visual perception was evaluated using the physical IMX219 camera and two-axis head.

A human face was presented at different locations within the camera field of view.

The detected face position generated corresponding pan and tilt adjustments, causing the physical camera head to follow changes in face location.

This experiment demonstrated a functional camera-to-actuator visual tracking loop.

Face following was evaluated independently from the final automatic voice-and-motion runtime and therefore does not establish simultaneous operation of every BUDDY modality.

G. Software-Level Verification

The movement and safety components were also evaluated using an implemented software test suite.

All 13 relevant tests completed successfully during development.

These results provide supporting evidence of expected software behaviour but are reported separately from physical experiments because successful software tests do not establish physical reliability under all hardware and environmental conditions.

VI. RESULTS AND DISCUSSION

Table III summarizes the principal physical observations.

TABLE III
PHYSICAL VOICE, MOVEMENT, AND SAFETY VALIDATION

Test	Test Condition	Observed Evidence
Wake-word range	0–3.0 m	Wake-word interaction remained usable within the tested range.
Extended voice range	Approx. 3.4 m	Recognition became inconsistent at approximately 3.4 m.
Forward	Spoken command	Robot executed forward physical movement through the safety-controlled pipeline.
Backward	Spoken command	Robot executed backward physical movement through the safety-controlled pipeline.
Left	Spoken command	Robot executed physical left-turn movement.
Right	Spoken command	Robot executed physical right-turn movement.
Stop	Spoken command	Active motor movement was stopped.
Obstacle stop	Approx. 0.10 m obstacle distance	Forward movement was interrupted by the safety layer when the obstacle was detected.
Obstacle removal	Obstacle removed	Active forward request was reevaluated and physical movement resumed.
Face following	Face position changed	Camera head physically followed the detected face using pan-tilt actuation.
Software safety	13 implemented tests	All 13 relevant software tests completed successfully.

The experiments demonstrate an end-to-end path from offline speech input to safety-mediated physical movement.

Successful recognition of a movement instruction did not itself authorize direct motor actuation. The request first passed through the independent safety supervisor.

The dynamic obstacle experiment provides the strongest evidence for this separation. The original forward request remained active while the physical environment changed.

Introducing an obstacle independently caused movement to stop, demonstrating that physical authorization was not fixed when the original voice request was accepted.

Removal of the obstacle allowed the still-active request to be reconsidered and movement to resume.

This behaviour demonstrates continuous environmental reevaluation rather than one-time obstacle checking at command initiation.

The camera experiment demonstrated a separate perception-to-actuation path in which visual observations influenced physical head orientation.

Together, these results demonstrate that auditory, environmental, and visual components can contribute complementary capabilities to the same embodied robotic platform.

A. Current Integration Boundaries

The reported experiments establish functional prototype feasibility rather than generalized statistical performance.

A controlled multi-participant speech study has not yet been completed; consequently, no generalized word-error rate or command-recognition accuracy is reported.

The seven-class facial-expression CNN has been developed, but complete live camera-to-expression-to-personalized-response operation has not yet been physically validated.

Dual-microphone acoustic localization remains experimental because reliable synchronized acquisition was not established with the current USB configuration.

Autonomous patrol, complete localization and mapping, and integrated facial identity recognition also remain continuing research extensions.

These boundaries distinguish physically demonstrated capabilities from developed or continuing components.

VII. LIMITATIONS AND FUTURE WORK

The current evaluation focuses on functional integration rather than generalized performance.

Future evaluation will include controlled multi-speaker speech trials, systematic end-to-end latency measurement, long-duration reliability testing, and power analysis.

The visual-affect component will be extended to combine facial-expression confidence, temporal persistence, recognized-user context, and interaction history before generating affect-aware responses.

This will enable comparison between conventional single-frame classification and a temporally stabilized personalized interaction approach.

The auditory research stream will investigate synchronized multi-channel audio hardware for acoustic source localization and audio-guided orientation.

The navigation research stream will be extended toward autonomous patrol, localization, mapping, route planning, and security-event response.

All future high-level autonomous capabilities will retain the existing architectural constraint that physical movement must pass through the independent safety-supervision layer.

VIII. CONCLUSION

This paper presented BUDDY, an intelligent multimodal robot-dog prototype integrating offline auditory interaction, visual perception, environmental sensing, and physical mobility using ROS 2.

The project combines three research streams involving auditory interaction, visual and facial intelligence, and environment-aware navigation and physical control.

A central system-level contribution is the separation of high-level movement intent from physical motor authorization.

An independent safety supervisor evaluates environmental observations before authorizing movement and continues to reevaluate active requests as environmental conditions change.

Physical experiments demonstrated automatic startup, offline voice-controlled movement, four-direction ultrasonic sensing, dynamic obstacle-mediated stopping, movement resumption after obstacle removal, and camera-based face following.

The principal safety experiment demonstrated that introducing an obstacle during an active forward request caused physical movement to stop, while removal of the obstacle allowed the still-active request to be reconsidered.

These results demonstrate the feasibility of coordinated multimodal human-robot interaction on embedded robotic hardware while preserving an explicit safety boundary between high-level intelligence and physical actuation.

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